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Roll no.

SECOND SEMESTER -- B.TECH
MID-SEMESTER EXAMINATION, MAY-2023

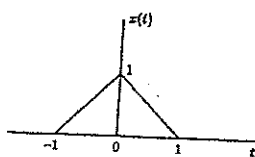
COURSE CODE: EEEEC03

COURSE TITLE: SIGNALS AND SYSTEMS

Time: 1.5 Hours

Max. Marks: 25

Note: Attempt All the five questions. Missing data information (if any), maybe suitably assumed.

Q.No.	Question	Marks	CO
Ques.1	Attempt all parts:		
1a.	Define and give the classification of continuous-time signals and discrete-time signals.	2.5	1
1b.	Prove that the power of the energy signal is zero over infinite time.	2.5	1
Ques.2	Attempt all parts:		
2a.	Check for linearity and time-invariance of the following system: $y(t) = t \cdot x(t)$	2	1
2b.	i. Determine if the following continuous-time systems are time invariant or time variant: $y(t) = x(t)$ ii. Differentiate between Static, Dynamic, Causal and Non-causal systems.	3	1
Ques.3	Attempt all parts:		
3a.	Explain the following in detail: i. Periodic and Non-periodic signal ii. Energy and Power signal	2	1
3b.	Explain time shifting, time scaling and time-reversal. A triangular pulse signal is shown in fig. 1. Sketch each of the following signals derived from $x(t)$. i. $x(3t)$ ii. $x(3t + 2)$ iii. $x(-2t - 1)$	3	2
 Fig. 1.			
Ques.4	Attempt all parts:		
4a.	Draw the following waveforms: i. $x_1(t) = u(t + 2)$ ii. $x_2(t) = \frac{5}{2}u(t - 2)$ iii. $x_3(t) = u(t) - 2u(t - 1) + u(t - 2)$ iv. $x_4(t) = \frac{d}{dx}x_3(t)$ v. $x_5(t) = x_1(t) - x_2(t)$	2.5	1
4b.	Draw the following waveforms: i. $x_6(t) = r(t) - r(t - 1) - u(t - 1)$ ii. $x_7(t) = u(t) + r(t) - 2r(t - 1) + r(t - 2) - u(t - 2)$ iii. $x_8(t) = -u(t + 1) - r(t + 1) - r(t - 1) - u(t - 1)$	2.5	2

	iv. $x_9(t) = r(t+2) - 2r(t+1) + 2r(t) - r(t-1) - u(t-3)$ v. $x_{10}(t) = r(t+2) - r(t) + r(t-2)$		
Ques.5	Attempt all parts:		
5a.	What is LTI system? Check the linearity of the system $y(t)$, $\frac{dy(t)}{dx} + y(t) = 5.x(t)$	2.5	2
5b.	What are causal systems? Check the causality of the following systems: i. $y(n) = x(-n)$ ii. $y(t) = x(t). \cos(t+1)$	2.5	2